

Project Report

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard

Project Type: Business Intelligence Dashboard

Industry: Manufacturing

Tool: Microsoft Power BI

Prepared By: Hemanth Chelimala

Version: 1.0

Date: July 2026

Executive Summary

Manufacturing organizations generate operational data across multiple departments including production, quality assurance, inventory, maintenance, and procurement. Managing these data sources independently makes it difficult for executives to obtain a unified view of organizational performance.

The Apex Manufacturing Executive Dashboard was developed to consolidate operational data into a single interactive Power BI solution. The dashboard provides decision-makers with real-time visibility into key manufacturing metrics through dynamic KPIs, analytical visualizations, and interactive filtering.

1. Project Overview

The objective of this project is to design and develop a business intelligence solution capable of monitoring manufacturing operations using Microsoft Power BI.

The dashboard integrates data from multiple business functions and transforms raw operational records into meaningful insights that support strategic and operational decision-making.

2. Business Problem

Before implementation, manufacturing reports were distributed across different operational departments.

Key challenges included:

- Fragmented reporting
- Manual report preparation
- Limited operational visibility
- Delayed decision-making
- Difficulty comparing manufacturing plants
- Limited performance monitoring

These challenges reduced reporting efficiency and made business analysis more time-consuming.

3. Project Objectives

The project objectives were to:

- Develop a centralized executive dashboard.
 - Monitor manufacturing performance.
 - Track production efficiency.
 - Analyze quality metrics.
 - Monitor maintenance activities.
 - Track inventory levels.
 - Evaluate procurement performance.
 - Improve business reporting.
-

4. Dataset Overview

The project uses a structured manufacturing dataset consisting of both transactional and master data.

Business entities include:

- Plants
- Products
- Employees
- Machines
- Production
- Quality
- Inventory
- Suppliers
- Purchase Orders
- Maintenance

These datasets provide the foundation for business analysis and KPI reporting.

5. Data Preparation

Data preparation was performed using Microsoft Power Query.

Activities included:

- Data import
- Data type validation
- Column standardization
- Relationship preparation
- Business rule validation
- Data quality review

Proper data preparation ensures reliable reporting and accurate business calculations.

6. Data Modeling

The dashboard follows a Star Schema architecture.

The model separates:

Dimension Tables

- Plants
- Products
- Employees
- Machines
- Suppliers

Fact Tables

- Production
- Quality
- Inventory
- Purchase Orders
- Maintenance

This structure improves performance and simplifies analytical calculations.

7. Dashboard Development

The Power BI solution contains six report pages.

Executive Dashboard

Provides an overview of organizational performance through executive KPIs.

Production Analysis

Analyzes production quantity, targets, and manufacturing efficiency.

Quality Analysis

Monitors inspection results, defect trends, and quality performance.

Maintenance Analysis

Tracks maintenance activities, machine utilization, and maintenance costs.

Inventory Analysis

Monitors stock availability and inventory performance.

Procurement Analysis

Analyzes supplier activity, procurement costs, and purchasing trends.

8. Key Performance Indicators

The dashboard monitors the following KPIs:

- Total Production
- Production Target
- Production Efficiency
- Defective Units
- Defect Rate
- Inventory Stock
- Maintenance Cost
- Purchase Orders
- Procurement Cost
- Active Suppliers

Each KPI dynamically responds to report filters and slicers.

9. Dashboard Features

The solution includes several interactive Power BI capabilities.

Features include:

- Dynamic KPI Cards
- Interactive Slicers
- Cross Filtering
- Responsive Charts

- Drill-down Analysis (where implemented)
 - Trend Analysis
 - Executive Reporting
-

10. Business Value

The dashboard provides significant value by enabling:

- Centralized reporting
 - Improved operational visibility
 - Faster decision-making
 - Better production monitoring
 - Enhanced inventory management
 - Maintenance cost tracking
 - Procurement analysis
 - Executive performance monitoring
-

11. Technical Skills Demonstrated

This project demonstrates practical knowledge of:

- Microsoft Power BI
 - Power Query
 - Data Modeling
 - Star Schema Design
 - DAX Measures
 - Data Visualization
 - KPI Development
 - Dashboard Design
 - Business Analytics
 - Manufacturing Analytics
-

12. Analytical Insights

The dashboard enables users to identify insights such as:

- Production performance across manufacturing plants.
- Achievement against production targets.
- Quality trends and defect monitoring.
- Inventory availability and stock movement.
- Maintenance expenditure and equipment performance.
- Procurement activity and supplier contribution.

The specific findings depend on the filters and reporting period selected by the user.

13. Business Recommendations

Based on dashboard analysis, organizations can:

- Improve production planning.
 - Monitor manufacturing efficiency.
 - Reduce product defects.
 - Optimize inventory levels.
 - Schedule preventive maintenance.
 - Evaluate supplier performance.
 - Reduce procurement costs.
 - Improve operational decision-making.
-

14. Future Enhancements

Potential future improvements include:

- Live ERP integration
 - Automated data refresh
 - Predictive analytics
 - Demand forecasting
 - Predictive maintenance
 - Row-Level Security (RLS)
 - Mobile dashboard optimization
 - Power BI Service deployment
 - Real-time manufacturing monitoring
-

15. Challenges and Learning Outcomes

During this project, several practical Power BI concepts were applied, including data preparation, relationship modeling, KPI development, dashboard design, and interactive reporting.

This project strengthened understanding of:

- Business intelligence workflow
 - Data transformation
 - Data modeling
 - DAX calculations
 - Dashboard storytelling
 - Manufacturing analytics
-

16. Project Deliverables

The completed project includes:

- Power BI Dashboard (.pbix)
 - Manufacturing Dataset
 - Dashboard Screenshots
 - Business Requirements Document
 - Data Dictionary
 - Data Profiling Report
 - Data Cleaning Report
 - Data Model Documentation
 - KPI Documentation
 - Dashboard User Guide
 - Project Report
 - GitHub Repository
-

17. Conclusion

The Apex Manufacturing Executive Dashboard demonstrates the complete lifecycle of a Business Intelligence project—from understanding business requirements and preparing data to designing a scalable data model and delivering interactive dashboards.

By combining manufacturing data from production, quality, inventory, maintenance, and procurement into a unified Power BI solution, the project enables stakeholders to monitor key operational metrics, identify performance trends, and make informed decisions.

This project showcases practical skills in Microsoft Power BI, Power Query, DAX, data modeling, and business analytics, making it a comprehensive portfolio project suitable for demonstrating data analyst capabilities.